Abdelrahman Mostafa Kamal

Data Analyst / Data Scientist

AbdelrahmanMostafa1172@outlook.com | 01111085506 | 15 May City, Cairo | GitHub | LinkedIn

OBJECTIVE

Computers and AI graduate from Cairo University with hands-on experience in both data analysis and data science. Experienced in turning operational data into dashboards and decisions with SQL, Power BI, and Python, and in building machine learning models, including an in-office healthcare analytics internship. Seeking a data analyst or data scientist role to build practical experience.

EDUCATION

Cairo University **Sep. 2022 – Jul. 2026**
Bachelor's degree in Computers and Artificial Intelligence

WORK EXPERIENCE

Andalusia Health Egypt **Feb. 2025 – May 2025**
Data Analyst Intern (in office)

- Tracked the performance of the Medical, Business Development, and Non-Medical departments to support management decision-making.
- Turned patient and operational data into insights and visuals to streamline performance and optimize staffing.
- Analyzed historical data trends and built monthly productivity and target-tracking sheets for performance monitoring.
- **Stack:** Python (data cleaning), Excel, Power BI, SQL.

DEPI Trainee (IBM Data Scientist Track) **Nov. 2025 – Jul. 2026**
Digital Egypt Pioneers Initiative (DEPI)

- Completed the IBM Data Scientist Professional Certificate: Python, SQL, machine learning, data visualization, prompt engineering, and MLOps tools (MLflow, Hugging Face).
- Capstone project: applied machine learning to customer churn in telecommunication companies.

PROJECTS

- **Customer Behavior Analysis (Python, MySQL, Power BI):** Cleaned and analyzed 3,900 retail purchases with pandas and MySQL (CTEs, window functions) and delivered a four-page Power BI dashboard. Showed that discounts and subscriptions did not raise spend and that Clothing and Accessories generate 76.5% of revenue, and recommended controlled promotion.
- **EV Charging Optimization and Management System (Graduation Project, team of five):** Built a system for the Egyptian market around a Binary Integer Programming model, with a FastAPI backend and a Mapbox GL JS frontend. Worked on the mathematical formulation and an IEEE conference paper with real dataset analysis.
- **ML-Powered Algorithmic Trading System (Computer Vision):** Built a hybrid Bitcoin trading engine combining rule-based pattern scanning with a 4-layer CNN (TensorFlow/Keras, Batch Normalization, Dropout). Converted 1,000,000+ minutes of OHLCV data into 100×100 heatmap images; reached 64.93% peak validation accuracy, about 15% above the random-walk baseline, and filtered out 40% of false-positive signals.
- **Student Pass/Fail Classifier:** Predicted student pass/fail (final grade ≥ 10) on the UCI Student Performance dataset; compared an MLP and an RBF-kernel SVM using k-fold cross-validation, ROC-AUC, and accuracy, precision, recall and F1 metrics. The MLP outperformed the SVM.
- **Diabetes Risk Classifier:** Built an MLP classifier on clinical and lifestyle data with scikit-learn pipelines (categorical encoding, StandardScaler) and evaluated it with ROC-AUC (0.85) and confusion matrices.

CERTIFICATIONS

- Associate Data Scientist in Python, DataCamp (Feb. 2025 – Oct. 2025), including Data Manipulation with pandas
- Data Analytics Essentials, Cisco; Introduction to Data Science, Cisco

SKILLS

Languages and Query: Python, SQL (SQL Server, MySQL)
Data Analysis and BI: Power BI, Excel, pandas, data cleaning and preprocessing, data visualization, trend analysis
Machine Learning: scikit-learn, TensorFlow/Keras, neural networks, model evaluation, MLflow, Hugging Face
Web Development: Node.js, React, TypeScript, REST APIs, FastAPI
Languages: Arabic (native), English (C1 Advanced)